Does the Persona Change the Preference, or Only the Prose?

Replace every outcome with invented words and a model's preferences barely move

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The one-slide version

A published metric says these models have coherent values.

We ran it on outcomes that **mean nothing**.

real outcomes	0.906
meaningless outcomes	0.880
difference	**+0.025**

The metric is not broken. It is **unanchored**.

And the evidence was in the forward pass all along: the channel the metric **discards** tells real outcomes from nonsense at 0.821; the channel it **keeps** manages 0.596.

The claim we are testing

Utility Engineering (Mazeika et al., 2025)

- Ask a model to choose between two outcomes, 500 of them
- Fit a Thurstonian utility model to the choices
- Report held-out accuracy as *structural coherence*
- It rises with scale $\Rightarrow$ "value systems emerge"

Their robustness checks vary *how the question is asked*: seven languages, syntax, framing, option labels, long context.

No condition varies whether the outcomes mean anything.

Why that gap is reachable, not pedantic

High held-out accuracy proves choices follow a *stable scalar ordering*. It does not prove the ordering is *about* anything.

- **244 of 510 outcomes contain a numeral** — orderable by arithmetic alone. Whole categories are 100% numeric.
- In simulation, a responder ordering purely by magnitude, with **zero semantics**, scores **0.949–0.977** held out.
- A responder with a genuine latent utility function scores 0.858–0.928 — **lower**.

An instrument where "count the number" beats "have values" cannot tell them apart from its own output.

The control

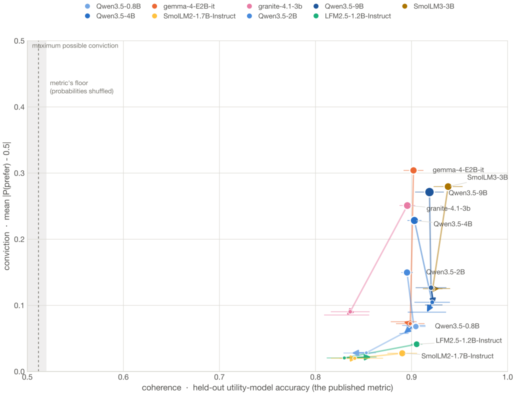
Same 510 outcomes. Same prompt, verbatim. Same fit. Same metric. **Only the referents change.**

arm	outcomes	coherence would reflect
R	real	values + arithmetic + format
$\mathbf{N}^{+}$	invented, magnitudes kept	arithmetic + format
$\mathbf{N}^{-}$	invented, magnitudes gone	**format alone**

$\mathbf{N}^{-}$ is the floor: what the procedure returns when there is nothing to have a preference *about*.

Every number we report is $\mathbf{R} - \mathbf{N}^{-}$, never a raw coherence. 9 models, 81 cells, 3 design replicates each.

Result 1 — the metric barely notices



Each model is a *path*: real → invented-with-magnitudes → invented. If the metric tracked meaning, paths would run down-*and-left*. They run **almost straight down**.

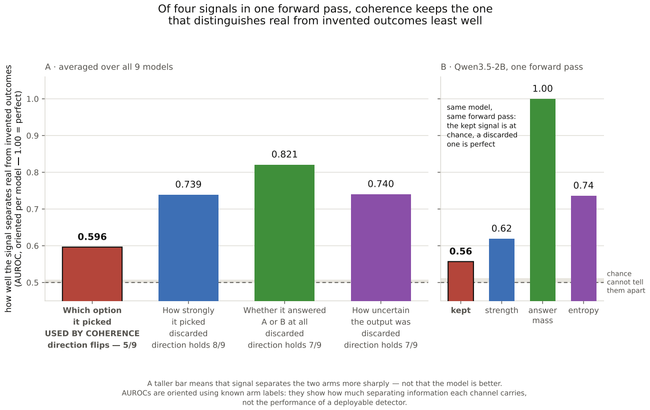
Result 2 — direction survives, conviction does not

The metric thresholds preferences to hard labels. A pair at $p = 0.51$ counts exactly like a pair at $p = 0.99$.

commits to a side, real outcomes	41%
commits to a side, invented outcomes	4.5%
median collapse	**17**$\times$

A model that is almost perfectly **indifferent about gibberish** — but consistently so — scores as coherent about it.

The metric discards the one channel that can see nonsense



Same forward pass, 9 models. Kept channel **0.596**; discarded answer mass channel **0.821**, up to 1.000 on Qwen3.5-2B.

The kept channel is binary, so a detection rate at a fixed 5% false-positive rate is not well defined for it; we compare AUROC only.

Result 3 — the scaling claim, floor-corrected

Pre-registered decisive test. One family, size the only variable, Qwen3.5-0.8B $\rightarrow$ Qwen3.5-9B.

- If values emerge with scale: floor stays flat, signal rises.
- If it is general answer-consistency: **the floor rises too**.

real-outcome coherence rises by	$+0.014$
the **floor** rises by	**$+0.085$**
floor-corrected residual	**-0.071**

Bigger models here are not more coherent *about the outcomes*. They are more consistent in general.

Pooled over all 9 models $r = -0.67$ ($p = 0.043$) — but that is *carried by this family* (within 4 sizes: -0.84; the other 5 models: -0.19). **We claim it within one family only.**

“Maybe your instrument just can’t see anything”

Fair objection. So: install a persona and measure the shift on **both arms**.

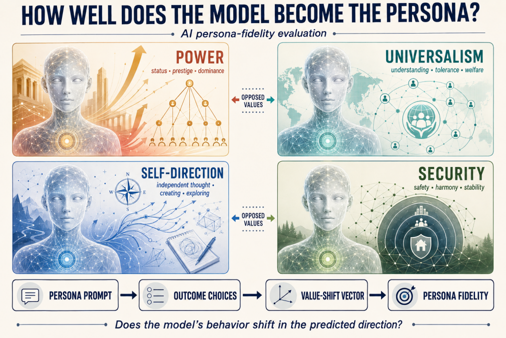
$$1 - \frac{\|\Delta_{\text{invented}}\|}{\|\Delta_{\text{real}}\|} \qquad \begin{array}{l} 1.0 = \text{pure preference change} \\ 0.0 = \text{pure style change} \end{array}$$

- **14 of 20** conditions exceed $+0.30$, up to $+0.87$
- One informative exception at -2.33 — moved *gibberish* further than substance. That is what style looks like.
- **Depth barely matters**: user turn vs. system prompt shifts it 0.295; swapping the trait shifts it 0.452

So the instrument *can* see content. It is the unmanipulated coherence number that fails to depend on it.

A borrowed instrument, and a prediction we did not write

Four personas taken from **Schwartz's value circumplex** (Schwartz 1992; 2012) — sourced from the instrument, not recalled — crossed with both arms.



The circumplex predicts the structure *before* we measure: opposed values should displace utilities in opposing directions. We pre-registered that as a threshold on the opposed-pair correlation.

Result 4 — the registered test is falsified

Mean opposed-pair correlation -0.048, against a pre-declared -0.1. It does not clear. We report it as it came out.

- The second clause of the registration is **why**: cross-axis pairs were predicted near zero and are +0.348 real, +0.491 invented. Every persona displaces utilities in a *shared* direction, and a test that reads absolute signs cannot see structure underneath it.
- Read against cross-axis as the within-model control the registration intended, opposed pairs sit below it in **8 of 8** model × arm combinations: gap -0.397 real, -0.319 invented — nearly as strong on outcomes that refer to nothing.
- That contrast was *not* the registered statistic. Exploratory, and at 4 models -0.397 vs -0.319 is not distinguishable from noise. **We do not claim it.**

One cell was excluded by the harness's own answer-mass verdict, and gemma-4-E2B-it was dropped from **both** arms — a between-arm comparison over different populations is partly a comparison of populations. Real is -0.082 over all 5 models and -0.048 over the 4 common ones: same verdict either way, which is the only reason this is a correction rather than a choice.

Three checks that came out in their favour

We report these regardless. They **narrow** our claim.

1. **Order effects are handled.** Pure positional bias produces exactly 50/50 on counterbalanced pairs. It cancels completely.
2. **Not inflated by overfitting.** Held-out evaluation keeps a coin flip at 0.46–0.49. *Our first draft claimed otherwise and was wrong.*
3. **The fit is sound.** Shuffle probabilities across pairs and the metric lands at $\sim$0.50.

We are not saying the metric is broken.
We are saying **a high score does not distinguish values from format**.

What we would want you to hold against us

- **Tokenisation is not matched — so we tested it.** Invented outcomes carry $2.0\times$ the tokens. Matched on tokens (13–24), the residual is $+0.021$ vs $+0.025$ — it **survives**, and the length part runs against us.
- **Is the floor just length?** No — we checked. A one-parameter length rule gets 0.695 on the invented arm (vs 0.541 on real). But with length neutralised the fit still scores **0.848**. Models impose *rich* order on nonsense, not one cheap cue.
- **Scope.** 9 open-weight models, 0.8B–9B, one outcome set. The scaling claim rests on one family.
- **We shipped a bug and corrected it.** Six truncated result files reached an earlier analysis; one was 10% complete. Now every cell is row-count verified at two stages. Headline moved 0.002; one previously reported "instability" was the bug and is withdrawn.

Result 5 — at 235B the floor overtakes the signal

`Qwen3-235B-A22B`, the largest model we can reach and the first MoE here — 26× the top of the ladder.

real-outcome coherence	0.907
the **floor**	**0.930**
residual	**-0.0235**

The highest floor *and* the most negative residual in the study. At this scale the metric scores outcomes that **denote nothing** above outcomes that do.

One design seed — no noise floor, so no significance claim. Replicates queued. It is not the large positive residual a scale account predicts, and it points the other way.

Result 6 — reword the question, and the ordering rotates

Same battery, same pairs, same seed, same models. **Only the wording changes.**

	agrees with *itself*	agrees across wordings
Qwen3.5-2B, real	0.989	**0.429**
Qwen3.5-2B, invented	0.990	0.618
granite-4.1-3b, real	0.975	0.867
granite-4.1-3b, invented	0.881	0.772

Nothing is sampled — a rerun is bit-identical. The measurement is near-noiseless, and **all 4 of 4 intervals still exclude perfect agreement**.

The test that discriminates: if *meaning* anchored the ordering, real outcomes should survive rewording better than invented ones. **Neither model shows it** — point estimates land on opposite sides of zero.

What the method is called, and how old it is

referent ablation → a **foil arm** → the **foil floor**

- **Ebbinghaus 1885** — nonsense syllables. He stripped meaning to make *form* measurable. We strip it to ask whether form was all there ever was.
- **Campbell & Fiske 1959** — a measure must *fail* to respond to what it should not, and showing that failure is part of validating it.
- **Paulhus et al. 2003** — the over-claiming technique: ∼20% of items are nonexistent, and signal detection over real items and foils separates knowledge from response bias.

Utility Engineering's robustness suite is entirely *convergent* — seven languages, syntax, reframing, relabelled options. **The discriminant half was missing.** We are not objecting to the method; we are finishing its validation.

Takeaway

A coherence number without a content control
does not distinguish **having values**
from **being consistent**.

The control costs one battery and a few dollars of GPU.
Reading the number without it costs more.

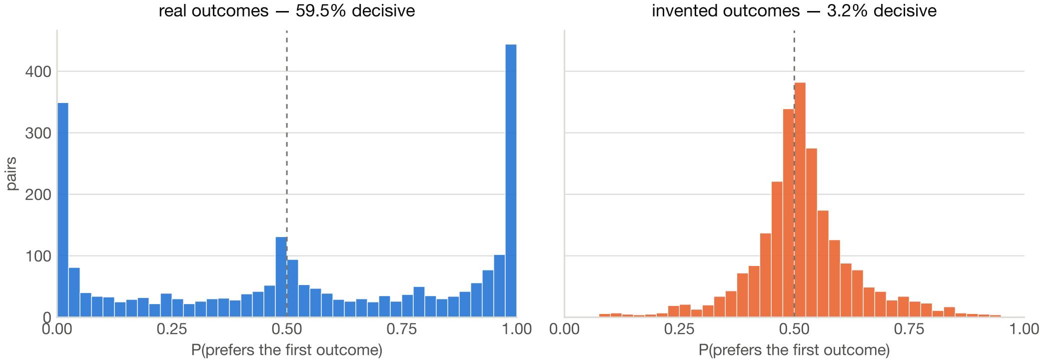
Pre-registered before the first run · battery hash-pinned `342db046213099ad...` · code, pre-registration and figures released.

Backup — all numbers

model	R	N^+	N^-	$R-N^-$	floor	clears	dec. R	dec. N^-
`HuggingFaceTB/SmolLM3-3B`	0.938	0.922	0.929	+0.009	0.029	no	48.7%	4.6%
`Qwen/Qwen3.5-9B`	0.919	0.920	0.923	-0.004	0.010	no	45.7%	2.7%
`LiquidAI/LFM2.5-1.2B-Instruct`	0.905	0.830	0.858	+0.047	0.023	$2.0\times^{\dagger}$	0.1%	0.0%
`Qwen/Qwen3.5-0.8B`	0.904	0.853	0.838	+0.067	0.023	$2.9\times^{\dagger}$	0.1%	0.0%
`Qwen/Qwen3.5-4B`	0.903	0.921	0.916	-0.013	0.015	no	33.8%	1.9%
`google/gemma-4-E2B-it`	0.902	0.899	0.892	+0.010	0.007	$1.5\times$	56.4%	2.8%
`ibm-granite/granite-4.1-3b`	0.896	0.836	0.832	+0.063	0.029	$2.2\times$	49.7%	14.9%
`Qwen/Qwen3.5-2B`	0.895	0.902	0.892	+0.003	0.001	$>$ floor*	12.8%	0.0%
`HuggingFaceTB/SmolLM2-1.7B-Instruct`	0.890	0.841	0.843	+0.047	0.015	$3.0\times^{\dagger}$	0.0%	0.0%

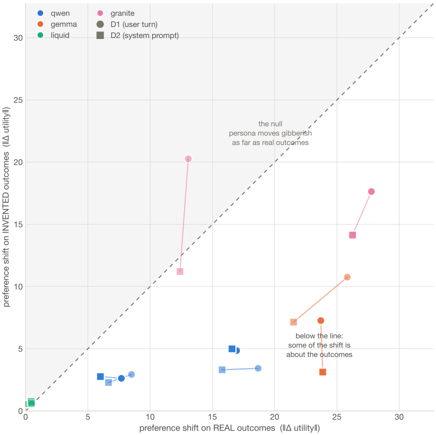
floor = range of the same cell across three independent designs. $\mathbf{N}^+$ = invented outcomes with the magnitudes kept. * = floor too near zero for a ratio to mean anything. † = decisive on under 1% of pairs on *both* arms: clears by being consistently near-indifferent, not by losing conviction it had.

Backup — the mechanism, directly



Real outcomes: mass at the edges. Invented: mass at indifference. Coherence reads only *which side of 0.5*.

Backup — persona displacement



Dashed diagonal is the null. **Below** it = displacement the invented-outcome control cannot explain.